

University of Mumbai

Examination 2020 under cluster 4 (Lead College: PCE)

Examinations Commencing from 23rd December 2020 to 6th January 2021 and from 7th January 2021 to 20th January 2021

Program: Computer Engineering

Curriculum Scheme: Rev/2016/

Examination: BE Semester VIII

Course Code: CSC802 and Course Name: Distributed Computing

Time: 2 hour

Max. Marks: 80

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Q1.	Choose the correct option for following questions. All the Questions are compulsory and carries equal marks. (2 marks each)
1.	The type of system in Distributed Computing is often _____
Option A:	Tightly Coupled System
Option B:	Loosely Coupled System
Option C:	Message Passing
Option D:	Data Parallelism
2.	What happens in a distributed computing when one of the site fails-
Option A:	The remaining sites can continue operating
Option B:	All the sites will stop working
Option C:	Directly connected sites will stop working
Option D:	All the running processes will be killed
3.	Which type of the transparency hides the failure and recovery of any resource?
Option A:	Relocation
Option B:	Replication
Option C:	Concurrency
Option D:	Failure
4.	The _____ communication is the form of communication that makes interaction possible among multiple processes running on different machines over the network.
Option A:	Intra-Process
Option B:	Inter-Process
Option C:	Task
Option D:	Process
5.	Which communication model uses the queuing model?

Option A:	MPI
Option B:	RPC
Option C:	MOM
Option D:	RMI
6.	An RMI skeleton is responsible for -
Option A:	Creating An Instance Of The Remote Object
Option B:	Exporting The Remote Object
Option C:	Binding The Instance Of The Remote Object To The RMI Registry
Option D:	Marshalling And Demarshalling
7.	In RPC, the parameters are _____.
Option A:	Pass By Parameter
Option B:	Pass By Reference
Option C:	Pass By Value
Option D:	Pass By Value And Pass By Reference
8.	The following is Physical Clock Synchronization Algorithm-
Option A:	Berkeley's
Option B:	Bully
Option C:	Lamport's Scalar Clock
Option D:	Vector Timestamp Ordering
9.	_____ is token based algorithm.
Option A:	Richart Agrawala
Option B:	Maekawa
Option C:	Lamport's
Option D:	Raymond Tree
10.	The time interval between critical region (CR) exit and new entry by any process is called
Option A:	System Throughput
Option B:	Message Complexity
Option C:	Synchronization Delay
Option D:	Response Time
11.	In distributed system, Election algorithms are based on the concept of-
Option A:	All processes are based on their IP address
Option B:	There is no priority number associated with any process
Option C:	Confidentiality
Option D:	A unique priority number is associated with each active process in system

12.	As per Lamport's clock synchronization algorithm, if timestamps of two events are same, then the events are _____
Option A:	S
Option B:	Non concurrent
Option C:	Synchronized
Option D:	Concurrent
13.	In _____ all the processes submitted by the users are distributed among the nodes of the system so as to equalize the workload among the nodes
Option A:	Virtualization
Option B:	Load Balancing
Option C:	Task Assignment
Option D:	Load sharing
14.	_____ determines when it will be necessary to transfer the process from one node to another
Option A:	Location policy
Option B:	Estimation policy
Option C:	Selection Policy
Option D:	Transfer Policy
15.	Migrating some resources, suspending the execution, migrating the process and then resuming is known as-
Option A:	Post Freezing
Option B:	Pre Transferring
Option C:	On demand Exchange
Option D:	Code migration
16.	What is not a desirable features of a good global scheduling algorithm?
Option A:	Flexible and scalable
Option B:	Ability to make dynamic scheduling decisions
Option C:	Apriori knowledge of process is required
Option D:	No apriori knowledge of process is required
17.	Which among these is a not a type of consistency models
Option A:	Sequential Consistency Model
Option B:	Processor Consistency Model
Option C:	Release Consistency Model
Option D:	Hybrid Model
18.	"The effect of a write operation by a process P on a data item X will always be seen by a successive read operation on X by the same process."
Option A:	Monotonic Reads

Option B:	Monotonic Writes
Option C:	Writes Follow Reads
Option D:	Read Your Writes
19.	Callback mechanism in AFS is used for managing-
Option A:	Backing up files
Option B:	RPC call to the server for file access
Option C:	Namespace of AFS
Option D:	File consistency among the local copies of the various clients using callback mechanism
20.	Which cache update policy copies every update to server at the same time?
Option A:	Write On Close
Option B:	Delayed Write
Option C:	Write Through
Option D:	Transaction Based

Q2	Solve any Two Questions out of Three. (10 marks each)
A	Explain RPC system model in detail.
B	Explain the distributed mutual algorithms for clock synchronization.
C	Discuss the issues in designing load balancing algorithm

Q3	Solve any Two Questions out of Three (10 marks each)
A	Discuss the need of the co-ordinator. Also give any one algorithm for co-ordinator selection.
B	Explain Data Centric and Client Centric Consistency models.
C	List desirable features of a good distributed file system. Explain file sharing semantics of it.